HW #5

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Question 1. Let $K = \mathbb{Q}(\sqrt[3]{3})$. The splitting of rational primes for K was done in Example 2.15.

(1) Show that the Galois closure of K over $\mathbb{Q}$ is $L := \mathbb{Q}(\sqrt[6]{-3})$. Namely, show that L is the smallest number field Galois over $\mathbb{Q}$ that contains K as a subfield. What is $\mathrm{Gal}(L/\mathbb{Q})$?

Hint. Use $\sqrt{3}e^{\pi i/6} = 2 + e^{2\pi i/3}$.

(2) Let $\alpha := \sqrt[6]{-3}$. Namely, let α be a root of the polynomial $X^6 + 3$. Compute

$$D(1, \alpha, \cdots, \alpha^5).$$

(3) Show that $[\mathcal{O}_L : \mathbb{Z}[\alpha]]$ is a power of 2.[1]

Hint. Use Exercise 1.7.

(4) Show that $\mathrm{disc}(L)$ is a power of 3.

Hint. Use that $L = \mathbb{Q}(\zeta_3, \sqrt[3]{3})$ and HW #2, Question 2.

(5) Let's say we take it for granted[2] that 2 is unramified in L. Let

$$(2) = \mathfrak{p}_1 \cdots \mathfrak{p}_g$$

be the prime ideal factorization of (2) in $\mathcal{O}_L$. Find g and the residue degrees of $\mathfrak{p}_1, \cdots, \mathfrak{p}_g$.

Hint. Use the computations in Example 2.15, and Exercise 2.4.

Proof. (1) Note that $\sqrt[3]{3}$ is a root of $X^3 - 3$, and the splitting field of $X^3 - 3$ is $\mathbb{Q}(\sqrt[3]{3}, \zeta_3\sqrt[3]{3}) = \mathbb{Q}(\sqrt[3]{3}, \zeta_3)$. We want to show that $\mathbb{Q}(\sqrt[6]{-3}) = \mathbb{Q}(\sqrt[3]{3}, \zeta_3)$. As both number fields have the matching degree (to be very precise, one has to show that $X^6 + 3$ is irreducible in $\mathbb{Z}[X]$, which is easy as the polynomial is Eisenstein at 3), it suffices to show that $\sqrt[6]{-3} \in \mathbb{Q}(\sqrt[3]{3}, \zeta_3)$.

Let us embed the number fields in $\mathbb{C}$, so that $\sqrt[3]{3}$ is identified with $\sqrt[3]{3}$ (the real cube root of 3) and ζ_3 is identified with $e^{2\pi i/3}$. We also identify $\sqrt[6]{-3}$ with $\sqrt[6]{3}e^{\pi i/6}$. By Hint,

$$\sqrt{3}e^{\pi i/6} = 2 + e^{2\pi i/3} = 2 + \zeta_3 \quad \rightsquigarrow \quad \sqrt[6]{3}e^{\pi i/6} = \frac{2 + \zeta_3}{\sqrt[3]{3}},$$

which implies that $\sqrt[6]{-3} \in \mathbb{Q}(\sqrt[3]{3}, \zeta_3)$, as desired. It is easy that $\mathrm{Gal}(L/\mathbb{Q}) \cong S_3$ (the permutation group in 3 elements).

(2) By Proposition 1.25,

$$D(1, \alpha, \cdots, \alpha^5) = (-1)^{\frac{6\cdot 5}{2}} N_{L/\mathbb{Q}}(6\alpha^5) = -6^6 N_{L/\mathbb{Q}}(\alpha)^5 = -6^3 \cdot 3^5.$$

(3) By (2), the possible prime factors of $[\mathcal{O}_L : \mathbb{Z}[\alpha]]$ are 2 and 3. However, as $X^6 + 3$ is Eisenstein at 3, by Exercise 1.7, 3 is coprime to $[\mathcal{O}_L : \mathbb{Z}[\alpha]]$. Thus, 2 is the only prime factor of $[\mathcal{O}_L : \mathbb{Z}[\alpha]]$, so it must be a power of 2.

[1] In fact, the Dedekind's criterion implies that $[\mathcal{O}_L : \mathbb{Z}[\alpha]] \neq 1$; can you see why?

[2] We will learn that this follows from the fact that $\mathrm{disc}(L)$ is not divisible by 2.

N.B.: If $\mathcal{O}_L = \mathbb{Z}[\alpha]$, then the Dedekind's criterion implies that (2) should factor as X^6+3 factors mod 2, but $X^6+3 \equiv X^6-1 \equiv (X^3-1)^2 \equiv (X-1)^2(X^2+X+1)^2 \pmod 2$, which contradicts the constancy of the residue degrees over the primes dividing 2.

(4) By HW #2, Question 2, the statement follows from the fact that both $\operatorname{disc}(\mathbb{Q}(\zeta_3))$ and $\operatorname{disc}(\mathbb{Q}(\sqrt[3]{3}))$ are $\pm$ powers of 3 (for $\mathbb{Q}(\zeta_3)$, either use $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ or Theorem 2.30(5); for $\mathbb{Q}(\sqrt[3]{3})$, this was done in Example 2.15).

(5) As $L/\mathbb{Q}$ is Galois, the residue degrees are the same. Let f be the residue degree. Then, we have $fg = [L:\mathbb{Q}] = 6$. Note that $\operatorname{Gal}(L/\mathbb{Q})$ is not abelian, so by Exercise 2.4, $f \neq 6$ (i.e., 2 cannot be inert in L). Thus, f can be either $1, 2, 3$. However, from Example 2.15, 2 is divisible by a prime $\mathfrak{q}$ in $K = \mathbb{Q}(\sqrt[3]{3})$ whose residue degree is 2. Consider $\mathfrak{q}\mathcal{O}_L$, which divides $2\mathcal{O}_L$, so $\mathfrak{q}\mathcal{O}_L$ is divisible by $\mathfrak{p}_i$ for some $1 \leq i \leq g$. This implies that $\mathfrak{p}_i \supset \mathfrak{q}\mathcal{O}_L$, so that $\mathfrak{p}_i \cap \mathcal{O}_K \supset \mathfrak{q}$. As $\mathfrak{q}$ is maximal, this implies that $\mathfrak{p}_i \cap \mathcal{O}_K = \mathfrak{q}$. This implies that

$$\mathcal{O}_K/\mathfrak{q} = \mathcal{O}_K/(\mathfrak{p}_i \cap \mathcal{O}_K) \hookrightarrow \mathcal{O}_L/\mathfrak{p}_i$$

Note that $\mathfrak{q}$ having residue degree 2 means that $\mathcal{O}_K/\mathfrak{q} \cong \mathbb{F}_4$. Thus, $\mathcal{O}_L/\mathfrak{p}_i$, which is isomorphic as a field to $\mathbb{F}_{2^f}$, is an extension of $\mathbb{F}_4$. Thus, f is even. This implies that f must be 2 and $g = 3$.

N.B.: We could have used that 2 is inert in $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ by using the same arguments as above.

□

Question 2. For a rational prime $p \equiv 5 \pmod 8$, this Question will find a unit[3] in $\mathbb{Q}(\sqrt{p})$ that is not itself a root of unity.

(1) Let

$$\alpha := \prod_{1 \leq a \leq p-1,\ a \text{ is a quadratic residue mod } p} (1+\zeta_p^a).$$

Show that $\alpha \in \mathbb{Q}(\sqrt{p})$.

(2) Show that α is a unit in $\mathbb{Q}(\sqrt{p})$. Namely, show that $\alpha \in \mathcal{O}_{\mathbb{Q}(\sqrt{p})}^\times$.

Hint. Use Exercise 1.5.

(3) Show that the only roots of unity in $\mathbb{Q}(\sqrt{p})$ are ± 1. Thus, our goal is to show that $\alpha \neq \pm 1$.

(4) There are two ways to embed $\mathbb{Q}(\sqrt{p})$ as a subfield of $\mathbb{R}$; $\sqrt{p} \in \mathbb{Q}(\sqrt{p}) \mapsto \sqrt{p} \in \mathbb{R}$ (the positive square-root of p) and $\sqrt{p} \in \mathbb{Q}(\sqrt{p}) \mapsto -\sqrt{p} \in \mathbb{R}$ (the negative square-root of p). Show that, in either way, α is sent to a positive real number. Deduce that $\alpha \neq -1$.

Hint. Use that $\left(\frac{-1}{p}\right) = 1$ so that one can divide quadratic residues mod p into pairs $\bigcup_{1 \leq a \leq \frac{p-1}{4},\ a \text{ is a quadratic residue mod } p} \{a, -a\}$.

(5) Using that $\left(\frac{2}{p}\right) = -1$, show that

$$\left(\prod_{1 \leq a \leq p-1,\ a \text{ is a quadratic residue mod } p} (1+X^a)\right) - 1$$

[3] We will learn later that there are infinitely many such units, and a more systematic way to find them.

is not divisible by $\Phi_p(X) = X^{p-1} + X^{p-2} + \cdots + 1$ (as an element in $\mathbb{Z}[X]$). Deduce that $\alpha \neq 1$.

Hint. Note that $\left(\frac{2}{p}\right) = -1$ implies that $2^{\frac{p-1}{2}} \equiv -1 \pmod{p}$.

Proof. (1) Note that, as $p \equiv 1 \pmod{4}$, the unique quadratic subfield of $\mathbb{Q}(\zeta_p)$ is $\mathbb{Q}(\sqrt{p})$. After identifying $\mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$, $(\zeta_p \mapsto \zeta_p^a) \leftrightarrow a$, $\mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}(\sqrt{p}))$ is the subgroup of quadratic residues mod p. It is obvious that α is fixed by any element of $\mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}(\sqrt{p}))$, which implies that $\alpha \in \mathbb{Q}(\sqrt{p})$.

(2) Note that $\alpha = N_{\mathbb{Q}(\zeta_p)/\mathbb{Q}(\sqrt{p})}(1 + \zeta_p)$. As $p \geq 5$, by Exercise 1.5, $1 + \zeta_p$ is a unit. Thus, α is also a product of units, which is itself a unit.

(3) One can embed $\mathbb{Q}(\sqrt{p})$ as a subfield of $\mathbb{R}$ (as is done in (4)), so any root of unity in $\mathbb{Q}(\sqrt{p})$ must be a root of unity in $\mathbb{R}$, but there are only two roots of unity in $\mathbb{R}$, namely, ± 1.

(4) Note that, for whatever embedding you choose, $(1 + \zeta_p^a)(1 + \zeta_p^{-a}) = 2 + 2\,\mathrm{Re}(\zeta_p^a) > 0$. Therefore, $\alpha > 0$ in whatever embedding you choose. Thus, $\alpha \neq -1$.

(5) Suppose not. Then, plugging $X = 1$, we get that $2^{\frac{p-1}{2}} - 1$ is divisible by $1 + 1 + \cdots + 1 = p$. However, by Hint, and as $p \neq 2$, we get a contradiction. This implies that ζ_p is not a root of $\left(\prod_{1 \leq a \leq p-1,\ a \text{ is a quadratic residue mod } p} (1 + X^a)\right) - 1$, so that $\alpha \neq 1$.

$\square$